



COMP9418 – Advanced Topics in Statistical Machine Learning

W6 - Sampling

Instructor: Edwin V. Bonilla

School of Computer Science and Engineering

(Last update 27/8/18 10:42 am)

Admin: Plan for the following weeks

- Week 7: Continuous latent variables (Amir)
- Week 8: Markov and Hidden Markov Models (Amir)
- Week 9: Undirected graphical models (Amir)
- 24/09/2018 - 30/09/2018 (Teaching recess, no teaching or tutorials)
- Week 10: NSW Public holiday (Labour day, 01/10/2018): No face-to-face lecture this week but a video will be recorded and posted on the course website. Everything else as usual.
- Week 11: I am back 😊

Acknowledgements

Material derived from:

- [\[Bishop, PRML, 2006\]](#) Pattern Recognition and Machine Learning, Christopher Bishop, 2006
 - Almost all figures in today's lecture are from this book
- [\[Murphy, MLaPP, 2012\]](#) Machine Learning: A Probabilistic Perspective, Kevin P. Murphy, 2012
 - Some figures taken from this book

Aims

This lecture will allow you to understand and apply stochastic approaches to probabilistic inference and estimation. Following it you should be able to:

- Sample from easy distributions where the cumulative distribution function and its inverse can be computed
- Understand and apply rejection sampling for drawing samples from low-dimensional distributions
- Understand and apply importance sampling approaches for computing expectations over functions
- Understand the fundamental theory and practicalities underlying Markov Chain Monte Carlo methods for sampling from desired distributions, including the Metropolis-Hastings algorithm and Gibbs sampling

Recall: The Need for Approximate Inference

- Computation of posterior (or expectations over it) is crucial:

$$p(\mathbf{Z}|\mathbf{X}) = \frac{\overset{\text{Likelihood}}{p(\mathbf{X}|\mathbf{Z})} \overset{\text{prior}}{p(\mathbf{Z})}}{\underset{\text{marginal}}{p(\mathbf{X})}}$$

- For many models exact computation is unfeasible
 - Intractability of marginal likelihood (sums or integrals in high dimensions)

Deterministic approaches

- Assumptions on (approximate posterior)
- Almost never exact
- Usually scalable
- *Variational inference*

Stochastic approaches

- **Sampling**
- Exact results given infinite computation
- Usually demanding, non-scalable

Inference via Numerical Sampling

- **Sampling**: drawing realisations $\mathcal{Z} = \{\mathbf{z}^{(\ell)}\}_{\ell=1}^L$ from $p(\mathbf{z})$
 - Techniques that rely upon these samples for estimation are known as Monte Carlo (MC) methods
- Posterior distribution over latent variables
 - Represented by a set of samples (unlike standard variational inference)
- **Estimate expectations** of a function using samples
- Computation of predictive distribution

$$p(y^{(\star)} | \mathbf{x}^{(\star)}, \mathcal{D}) = \int \underbrace{p(y^{(\star)} | \mathbf{x}^{(\star)}, \boldsymbol{\theta}) p(\boldsymbol{\theta} | \mathcal{D})}_{\text{Expectation}} d\boldsymbol{\theta}$$

function posterior

- MC can be used within other inference methods
 - Expectation maximisation
 - Variational inference

Estimating π via Monte Carlo Integration

Area of circle of radius r :

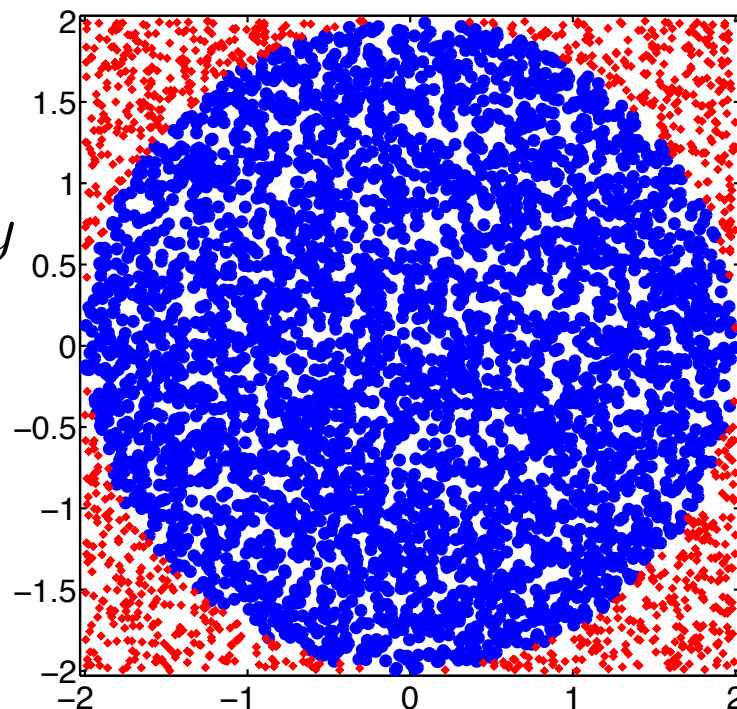
$$\pi r^2 = I = \int_{-r}^r \int_{-r}^r \underbrace{\mathbb{I}(x^2 + y^2 \leq r^2)}_{f(x, y)} dx dy$$

$$p(x) = p(y) = \text{Uniform}(-r, r) = \frac{1}{2r}$$

Then:
$$I = 4r^2 \int \int f(x, y) p(x) p(y) dx dy$$

$$\approx 4r^2 \frac{1}{S} \sum_{s=1}^S f(x^{(s)}, y^{(s)})$$

$$\pi \approx 4 \frac{1}{S} \sum_{s=1}^S f(x^{(s)}, y^{(s)})$$



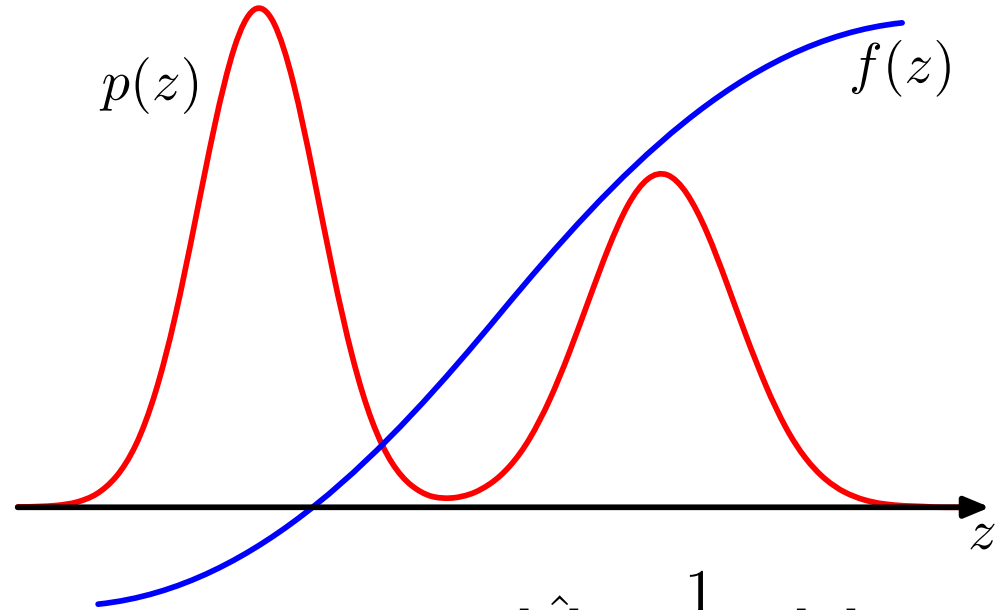
Estimating Expectations Using Samples

Given $\mathcal{Z} = \{\mathbf{z}^{(\ell)}\}_{\ell=1}^L$ drawn independently from $p(\mathbf{z})$

$$\mathbb{E}[f] = \int f(\mathbf{z})p(\mathbf{z})d\mathbf{z}$$

- Can be approximated with:

$$\hat{f} = \frac{1}{L} \sum_{\ell=1}^L f(\mathbf{z}^{(\ell)})$$



- Unbiased estimate $\mathbb{E}[\hat{f}] = \mathbb{E}[f]$ with variance $\mathbb{V}[\hat{f}] = \frac{1}{L} \mathbb{V}[f]$
- Variance Does not depend on dimensionality of $\mathbf{z}$
 - High accuracy with a small # samples
 - » However, samples may not be independent
 - » Effective sample size much smaller than apparent sample size
 - » Expectation may be dominated by regions of small probability

Why is Sampling Hard

We will usually assume that we can evaluate $p(\mathbf{z})$ at least to within a multiplicative constant, i.e. we can evaluate $\tilde{p}(\mathbf{z})$:

$$p(\mathbf{z}) = \frac{1}{Z_p} \tilde{p}(\mathbf{z})$$

First problem: We don't know Z_p , which can be hard to compute:

$$Z_p = \int \tilde{p}(\mathbf{z}) d\mathbf{z}$$

Second problem: Even if we knew Z_p :

- Problem still challenging, esp. in high-D spaces
 - How to identify those places where $p(\mathbf{Z})$ is big without evaluating it everywhere?

Outline

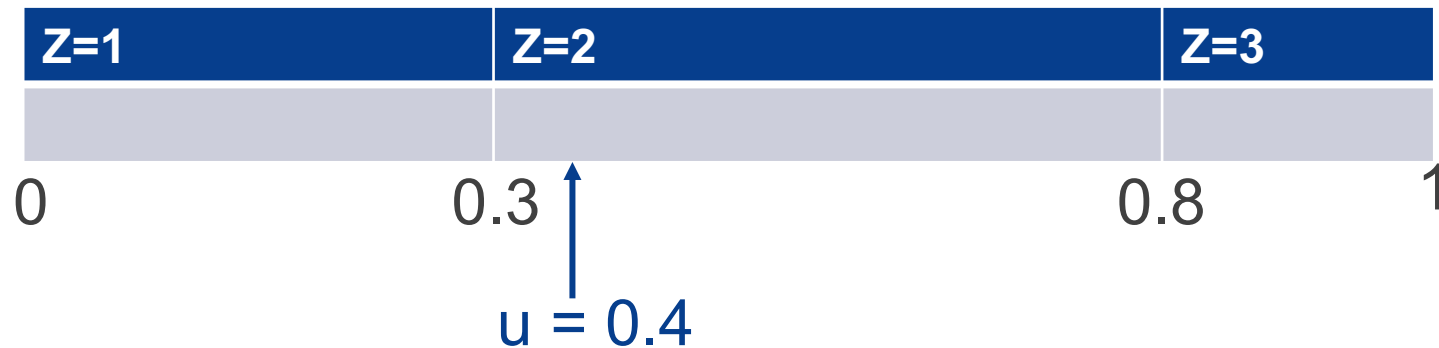
- I. Easy Sampling
- II. Rejection Sampling
- III. Importance Sampling
- IV. Markov Chain Monte Carlo
 - 1. Metropolis
 - 2. Metropolis-Hastings
 - 3. Gibbs Sampling

I. Easy Sampling

Sampling from Discrete Random Variables

Categorical Distribution

Let $z \in \{1, 2, 3\}$ and $p(z) = \text{Cat}(z|\pi)$ with $\pi = (0.3, 0.5, 0.2)$

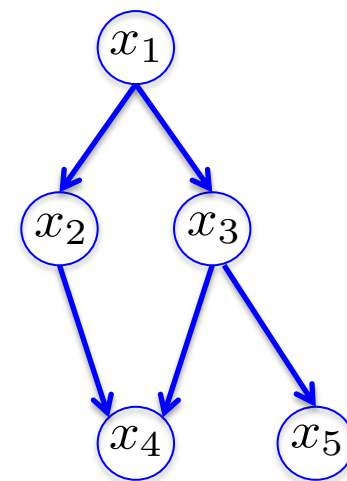


$u \sim \text{Uniform}(0, 1)$

$u = 0.4 \rightarrow z = 2$

Sampling from a Bayesian Network

$$p(\mathbf{z}_{1:D}) = \prod_{d=1}^D p(z_d | \mathbf{z}_{\pi(d)})$$



- Topological order, $\pi(d)$: Parents of node d
- **Ancestral sampling** without evidence
 - Sample the root node using corresponding Categorical distribution
 - Sample from the conditionals $z_d \sim p(z_d | \mathbf{z}_{\pi(d)})$
 - After one pass we have one sample from the joint $p(\mathbf{z})$
- With evidence
 - Accept / reject sample if they agree / disagree with evidence
 - » Reject sample as soon as a variable disagree with evidence
 - Inefficient and rarely used in practice
 - » Acceptance probability decreases rapidly with # observed variables

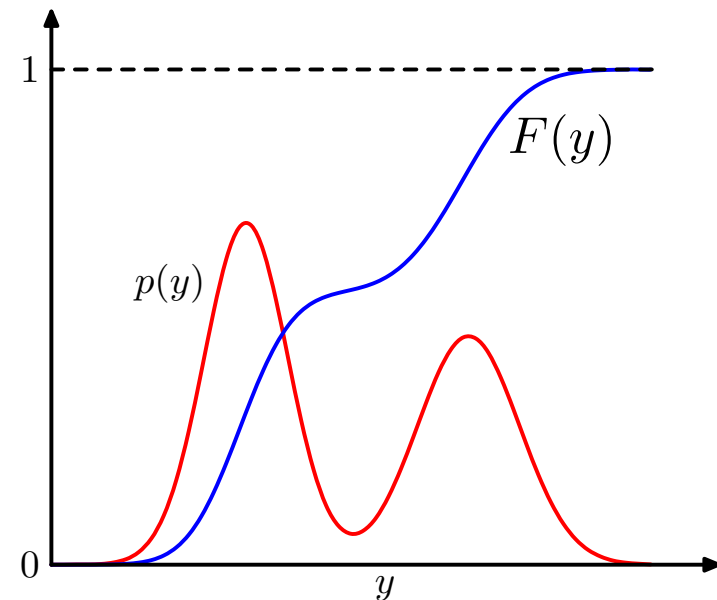
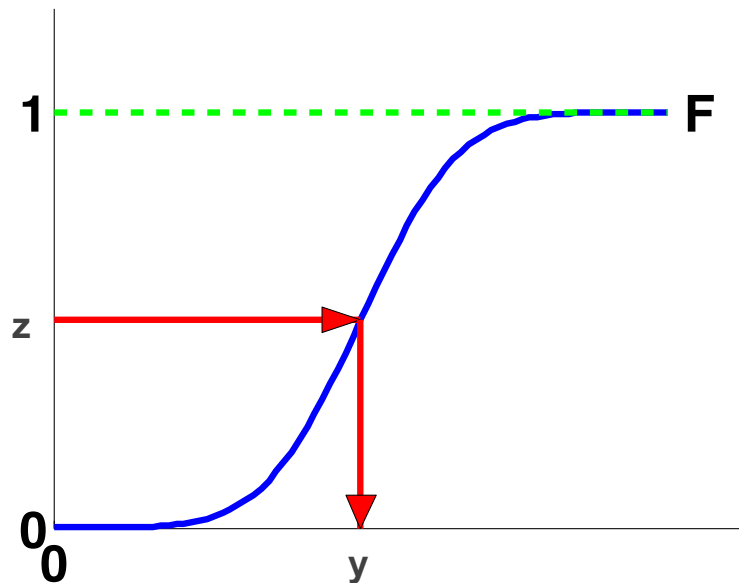
Standard Distributions: Using the Inverse Cdf

- Assume z uniformly distributed over $(0,1)$
- Transform z : $y = g(z)$

Cumulative distribution function (cdf)

- Then $p(y) = p(z) \left| \frac{dz}{dy} \right| \longrightarrow z = F(y) = \int_{-\infty}^y p(\hat{y}) d\hat{y}$

- Then $y = F^{-1}(z)$ Inverse cdf



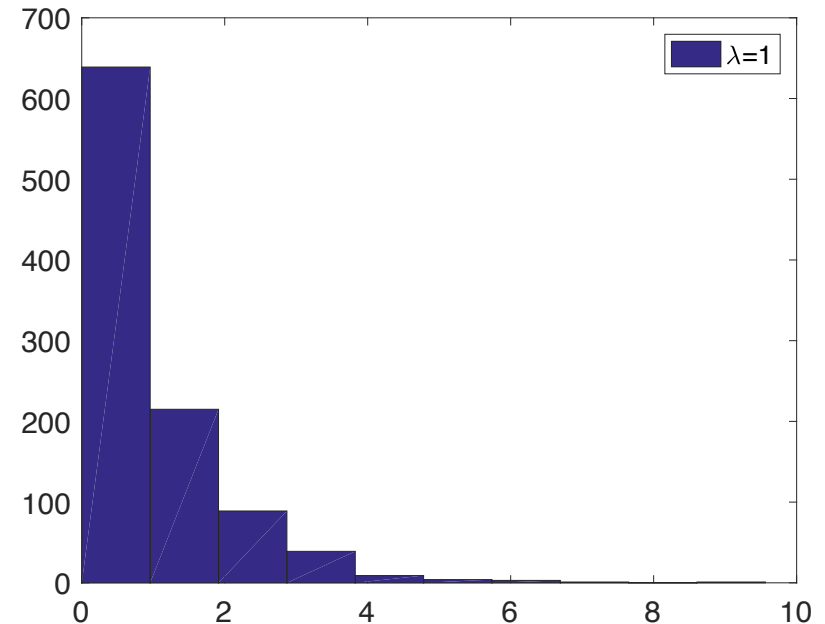
Example: The Exponential Distribution

Let $p(y|\lambda) = \text{Exponential}(y|\lambda) = \lambda \exp(-\lambda y)$ where $0 \leq y < \infty$

Then
$$F(y) = \int_0^y \lambda \exp(-\lambda \hat{y}) d\hat{y} = 1 - \exp(-\lambda y)$$

$$F^{-1}(y) = -\lambda^{-1} \log(1 - y)$$

1. $z \sim \text{Uniform}(0, 1)$
2. $y = -\lambda^{-1} \log(1 - z)$



Example: The Cauchy Distribution

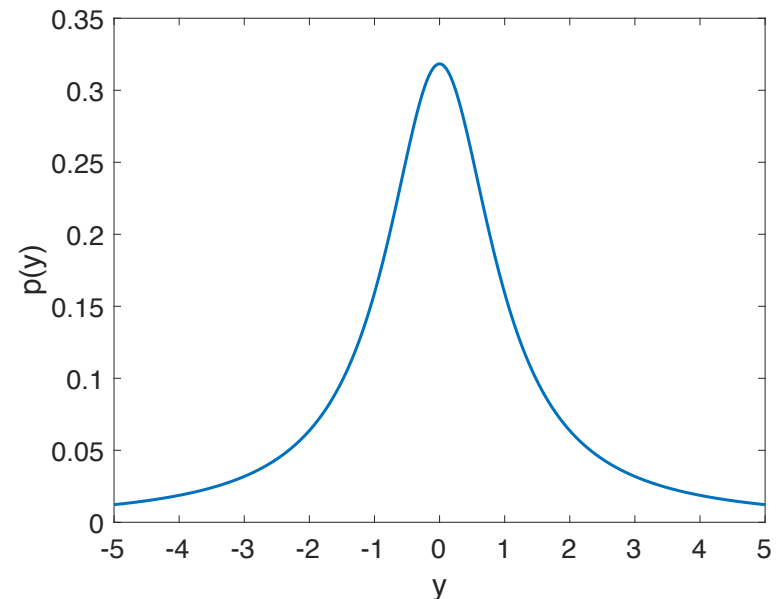
$$p(y|m, b) = \text{Cauchy}(y|m, b) = \frac{1}{\pi} \frac{b}{(y - m)^2 + b^2}$$

- The cdf is given by: $F(y) = \frac{1}{2} + \frac{1}{\pi} \tan^{-1} \left(\frac{y - m}{b} \right)$

$$F^{-1}(y) = m + b \tan \left[\pi \left(y - \frac{1}{2} \right) \right]$$

1. $z \sim \text{Uniform}(0, 1)$

2. $y = m + b \tan \left[\pi \left(z - \frac{1}{2} \right) \right]$



Generalization to Multiple Variables

$$p(y_1, \dots, y_D) = p(z_1, \dots, z_D) \left| \frac{\partial(z_1, \dots, z_D)}{\partial(y_1, \dots, y_D)} \right|$$

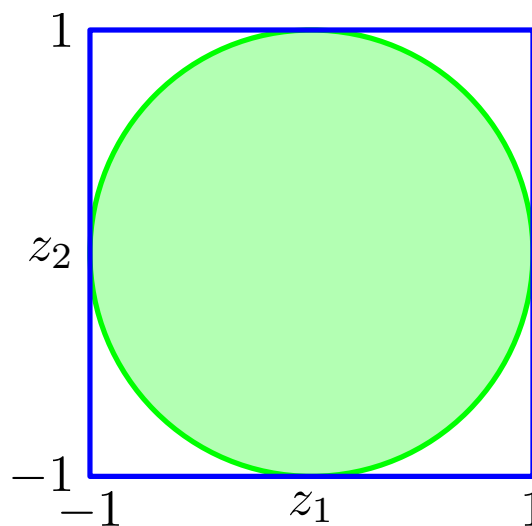
Jacobian
Determinant

Box-Muller method for generating samples from a Gaussian

1. z_1, z_2 uniformly over $(-1, 1)$ How?
2. Only retain samples if $z_1^2 + z_2^2 \leq 1$

→ Uniform distribution over the unit circle

$$p(z_1, z_2) = \frac{1}{\pi}$$



Sampling from a Gaussian (cont'd)

Let $r^2 = z_1^2 + z_2^2$

$$y_1 = z_1 \left(\frac{-2 \log z_1}{r^2} \right)^{1/2} \quad y_2 = z_2 \left(\frac{-2 \log z_2}{r^2} \right)^{1/2}$$

Then

$$\begin{aligned} p(y_1, y_2) &= p(z_1, z_2) \left| \frac{\partial(z_1, z_2)}{\partial(y_1, y_2)} \right| \\ &= \left[\frac{1}{\sqrt{2\pi}} \exp(-y_1^2/2) \right] \left[\frac{1}{\sqrt{2\pi}} \exp(-y_2^2/2) \right] \\ &= \mathcal{N}(y_1|0, 1) \quad \mathcal{N}(y_2|0, 1) \end{aligned}$$

y_1, y_2 : Samples from independent standard Gaussians

Sampling from a Correlated Gaussian

How to sample from $p(\mathbf{y}) = \mathcal{N}(\mathbf{y}|\boldsymbol{\mu}, \boldsymbol{\Sigma})$?

1. $\mathbf{z}$: Sample from a uncorrelated Gaussian, i.e: $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$

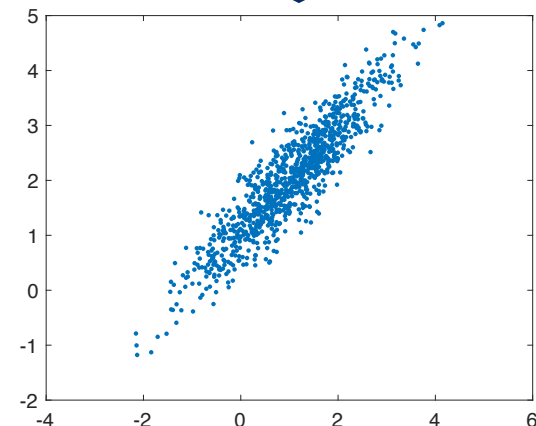
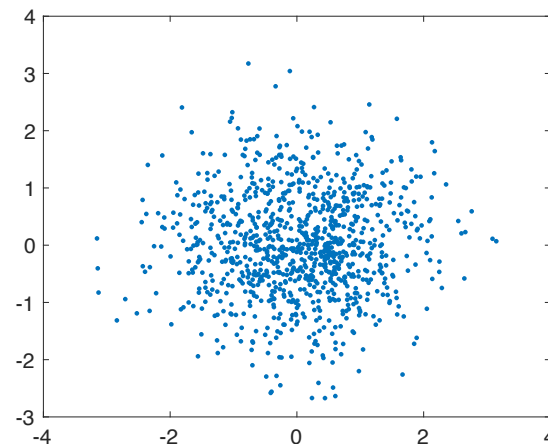
2. Decompose $\boldsymbol{\Sigma}$ such that $\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^T$
– Using the Cholesky decomposition

3. Make $\mathbf{y} = \boldsymbol{\mu} + \mathbf{L}\mathbf{z}$

- Then $p(\mathbf{y})$ is a Gaussian with

$$\mathbb{E}[\mathbf{y}] = \boldsymbol{\mu}$$

$$\text{Cov}[\mathbf{y}] = \mathbf{L}\mathbf{L}^T = \boldsymbol{\Sigma}$$



II. Rejection Sampling

Rejection Sampling (1)

- Inverse cdf method (or transformation method) relies on computation the integral and its inversion
 - Only feasible for limited number of simple distributions
- **Goal:** Sample from non-standard distribution $p(z)$
 - Assume we can evaluate $p(z)$ up to a normalisation constant

$$p(z) = \frac{1}{Z_p} \tilde{p}(z)$$

Known un-normalised version of $p(z)$

Unknown normalisation constant

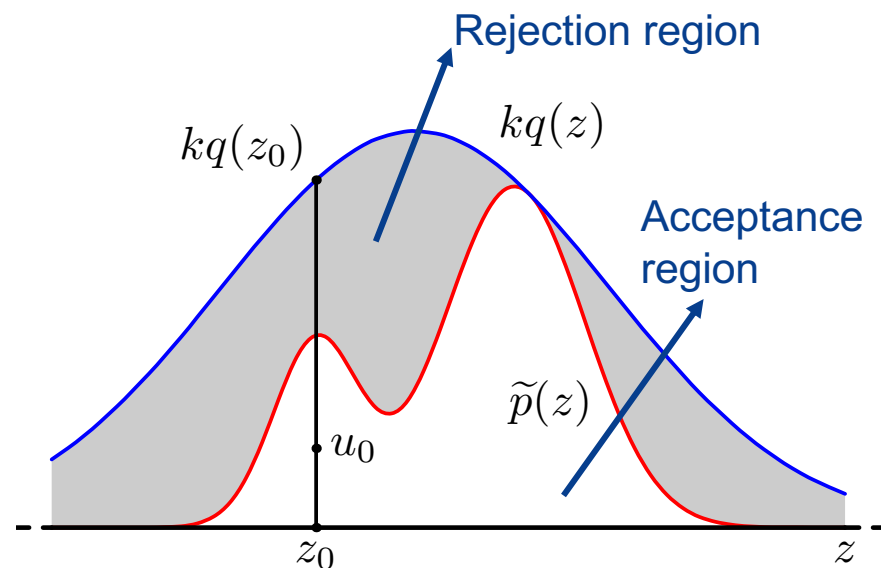
Application example?

- Let $q(z)$ be a (simpler) *proposal distribution (or envelope distribution)*, from which we can draw samples
- Choose k such that the comparison function $kq(z) \geq \tilde{p}(z)$

Rejection Sampling (2)

1. $z_0 \sim q(z)$
2. $u_0 \sim \text{Uniform}(0, kq(z_0))$
 - Pair (z_0, u_0) has uniform distribution under curve $kq(z)$
3. If $u_0 > \tilde{p}(z_0)$
 Reject sample
 else
 Accept sample

$$p(\text{accept}) = \int \underbrace{\frac{\tilde{p}(z)}{kq(z)}}_{p(\text{accept} | z)} \underbrace{q(z)}_{\text{Prob. of drawing } z} dz$$



- Retained points uniformly distributed under curve $\tilde{p}(z)$
 - Corresponding z values distributed $p(z)$ **Exercise**

Efficiency depends on k

→ As small as possible
subject to $kq(z) \geq \tilde{p}(z)$

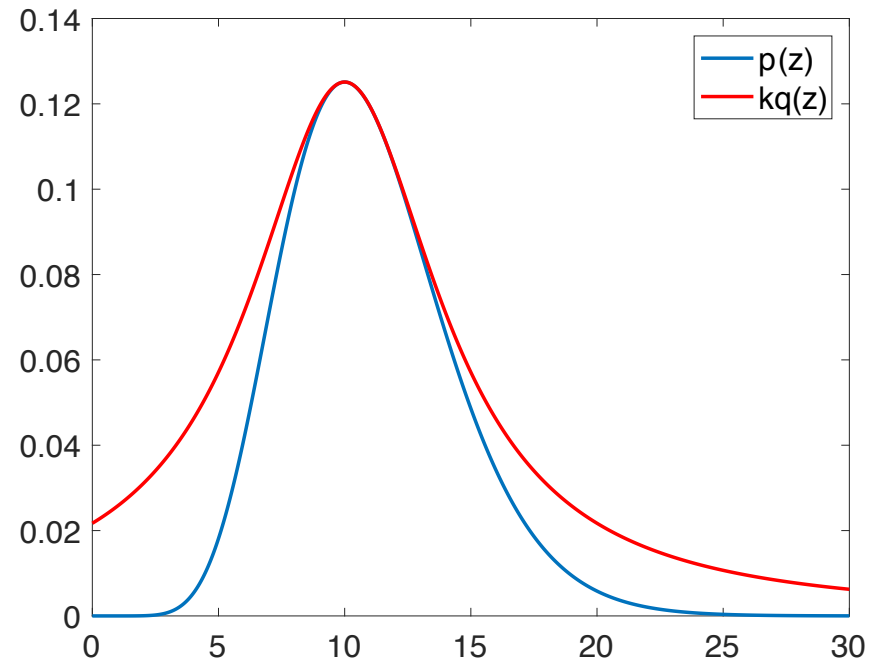
Rejection Sampling Example

Goal: Sample from $\underbrace{\text{Gamma}(z|\alpha, \beta)}_{p(z)} = \frac{1}{\Gamma(\alpha)} \beta^\alpha z^{\alpha-1} \exp(-\beta z)$

- $\alpha > 1 \rightarrow$ bell-shaped
- $q(z)$: Cauchy also bell-shaped and can sample from it
 - Using variable transformation

- $\alpha=11, \beta=1$

1. $y \sim \text{Uniform}(0, 1)$
 2. $z = m + b \tan y$
- Then $kq(z) = \frac{k}{1 + (z - m)^2/b^2}$
3. Set $m = a - 1$ $b^2 = 2a - 1$
 - Minimum k



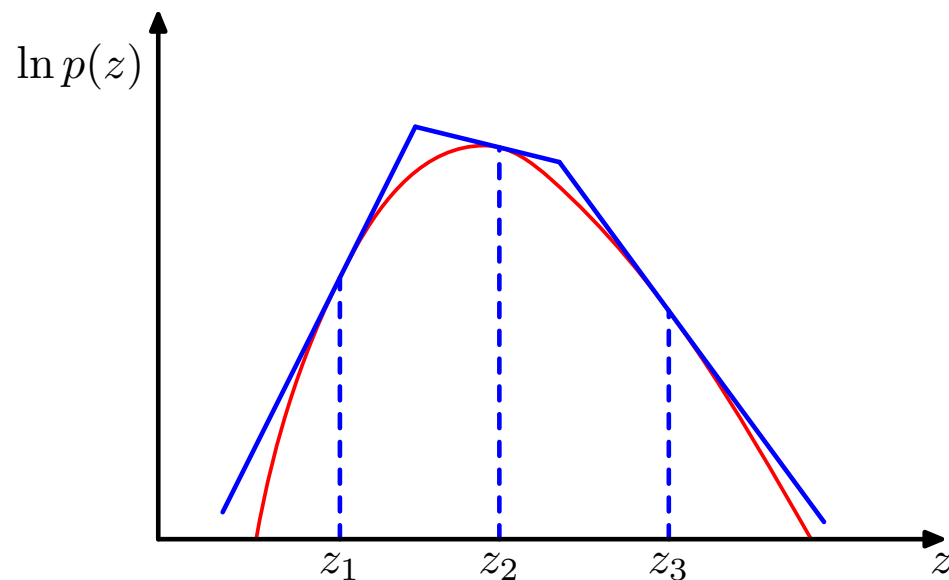
Application to Bayesian Statistics

- Goal: Draw samples from $p(\boldsymbol{\theta}|\mathcal{D}) = \frac{p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathcal{D})}$
- Target distribution: $\tilde{p}(\boldsymbol{\theta}) = p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})$
- Proposal: $q(\boldsymbol{\theta}) = p(\boldsymbol{\theta})$
- With $k = p(\mathcal{D}|\hat{\boldsymbol{\theta}}_{\text{ML}})$  $p(\text{accept}) = \int \frac{p(\mathcal{D}|\boldsymbol{\theta})p(\boldsymbol{\theta})}{p(\mathcal{D}|\hat{\boldsymbol{\theta}}_{\text{ML}})} d\boldsymbol{\theta}$
- Samples from prior with high likelihood $\rightarrow$ more likely to be retained in the posterior
 - Can be inefficient if big mismatch prior vs likelihood

Adaptive Rejection Sampling

- Determining analytical envelope distribution may be hard
 - We can construct it on the fly
 - Log concave distributions is easy
 - » Derivatives are not increasing functions of z
- Upper bound log-density with piecewise linear function

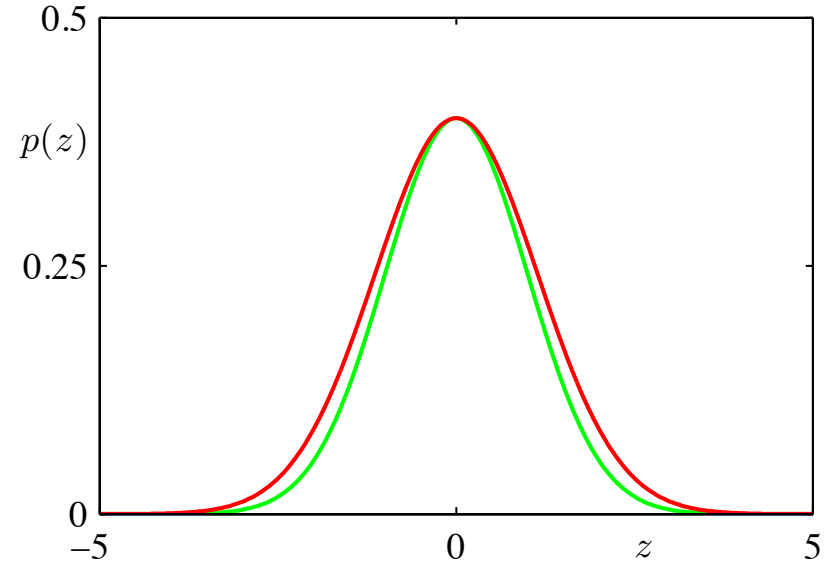
1. Initial locations on a grid
2. Build envelope using tangent lines
3. Draw sample from envelope (piecewise exponential)
4. Apply usual rejection criterion
 - If sample rejected, refine grid



Envelope becomes tighter as the # grid points increases

High Dimensionality and Other Issues

- Imagine we wish to sample from $\mathcal{N}(\mathbf{0}, \sigma_p^2 \mathbf{I})$ in D dimensions
- Use proposal $\mathcal{N}(\mathbf{0}, \sigma_q^2 \mathbf{I})$ with $\sigma_q^2 \geq \sigma_p^2$ so that $kq(z) \geq p(z)$
- Optimal $k = (\sigma_q/\sigma_p)^D$
- Acceptance rate $1/k$
- Diminishes exponentially with dimensionality!



- Finding good proposals and comparison functions is hard
 - E.g. for multi-modal distributions
- Useful in 1D and 2D and as a subroutine to other algorithms

III. Importance Sampling

Importance Sampling (1)

- Framework for approximating expectations, not for sampling
- Drawing samples from $p(\mathbf{z})$ is impractical but can evaluate it
- **Idea:** sample from proposal distribution $q(\mathbf{z})$
 - Weight samples with $p(\mathbf{z})/q(\mathbf{z})$

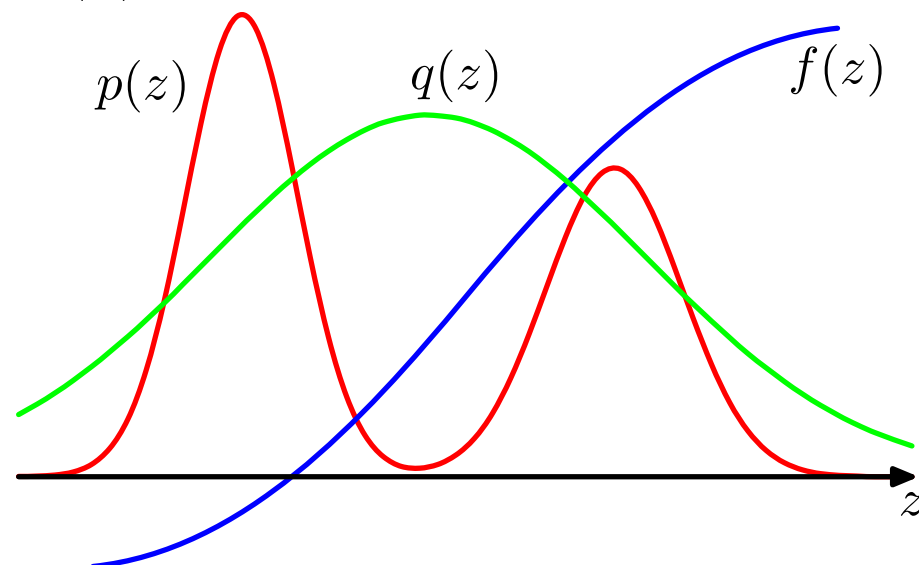
Expectation over $q(\mathbf{z})$

$$\mathbb{E}_{p(\mathbf{z})}[f(\mathbf{z})] = \int f(\mathbf{z})p(\mathbf{z})d\mathbf{z} = \int \frac{p(\mathbf{z})}{q(\mathbf{z})} f(\mathbf{z})q(\mathbf{z})d\mathbf{z}$$

$$\approx \frac{1}{L} \sum_{\ell=1}^L \boxed{\frac{p(\mathbf{z}^{(\ell)})}{q(\mathbf{z}^{(\ell)})}} f(\mathbf{z}^{(\ell)})$$

r_l : Importance weights

where $\{\mathbf{z}^{(\ell)}\}_{\ell=1}^L \sim q(\mathbf{z})$



Importance Sampling (2)

- Also works for unnormalised $\tilde{p}(\mathbf{z})$ and $\tilde{q}(\mathbf{z})$

$$\begin{aligned}\mathbb{E}_{p(\mathbf{z})}[f(\mathbf{z})] &= \int f(\mathbf{z})p(\mathbf{z})d\mathbf{z} = \frac{Z_q}{Z_p} \int \frac{\tilde{p}(\mathbf{z})}{\tilde{q}(\mathbf{z})} f(\mathbf{z})q(\mathbf{z})d\mathbf{z} \\ &\approx \frac{1}{L} \frac{Z_q}{Z_p} \sum_{\ell=1}^L \tilde{r}_\ell f(\mathbf{z}^{(\ell)}) \quad \text{where} \quad \{\mathbf{z}^{(\ell)}\}_{\ell=1}^L \sim q(\mathbf{z}) \quad \tilde{r}_\ell = \frac{\tilde{p}(\mathbf{z}^{(\ell)})}{\tilde{q}(\mathbf{z}^{(\ell)})}\end{aligned}$$

- And the ratio $\frac{Z_p}{Z_q} = \frac{1}{Z_q} \int \tilde{p}(\mathbf{z})d\mathbf{z} = \int \frac{\tilde{p}(\mathbf{z})}{\tilde{q}(\mathbf{z})} q(\mathbf{z})d\mathbf{z} \approx \frac{1}{L} \sum_{\ell=1}^L \tilde{r}_\ell$

- Hence $\mathbb{E}_{p(\mathbf{z})}[f(\mathbf{z})] \approx \sum_{\ell=1}^L w_\ell f(\mathbf{z}^{(\ell)})$ where $w_\ell = \frac{\tilde{r}_\ell}{\sum_{m=1}^L \tilde{r}_m}$

Importance Sampling: Remarks

- Note that, unlike rejection sampling, we use all the samples
- Success depends on how well $q(\mathbf{z})$ matches $p(\mathbf{z})$
 - Would like to choose sample points where $p(\mathbf{z}) f(\mathbf{z})$ is large
- Importance weights may be dominated by few very large weights
 - Effective sample size much smaller than apparent sample size
- Problems where samples outside regions where $p(\mathbf{z})f(\mathbf{z})$ large
 - Apparent variance is small even if expectation is completely wrong!
 - No diagnostics for this arbitrarily large error
- Key requirement: $q(\mathbf{z})$ should not be small where $p(\mathbf{z})$ large

Importance Sampling for DAGs

- When no evidence use ancestral sampling
- With evidence, rejecting inconsistent samples is inefficient

Likelihood weighting sampling

1. If variable in evidence, set it to its given value
 2. Otherwise, sample from its conditional distribution given its parents (which are set to their currently sampled values)
- Equivalent to using proposal: $q(\mathbf{z}) = \prod_{t \notin E} p(x_t | \mathbf{x}_{\pi(t)}) \prod_{t \in E} \delta_{x_t^*}(x_t)$

$$r(\mathbf{z}) = \frac{p(\mathbf{z})}{q(\mathbf{z})} = \prod_{t \notin E} \frac{p(x_t | \mathbf{x}_{\pi(t)})}{p(x_t | \mathbf{x}_{\pi(t)})} \prod_{t \in E} \frac{p(x_t | \mathbf{x}_{\pi(t)})}{1} = \prod_{t \in E} p(x_t | \mathbf{x}_{\pi(t)})$$

Weights are simply the likelihood of the evidence

Sampling-Importance-Resampling

- Success of rejection sampling depends on suitable k
 - Impractically very large values $\rightarrow$ Very small acceptance rates

1. Draw L samples $(\mathbf{z}^{(1)}, \mathbf{z}^{(2)}, \mathbf{z}^{(L)})$ from $q(\mathbf{z})$
2. Build weights w_i using importance sampling step
3. Draw second set of samples from the discrete distribution $(\mathbf{z}^{(1)}, \mathbf{z}^{(2)}, \mathbf{z}^{(L)})$ with probabilities $(w_1, w_2, \dots, w_L)$

Resulting samples are only approximately distributed according to $p(\mathbf{z})$ but correct as $L \rightarrow \infty$

IV. Markov Chain Monte Carlo (MCMC)

Basic Metropolis Algorithm (1)

- Rejection sampling and importance sampling severely limited
 - Especially in high dimensions
- **Goal:** A flexible algorithm that scales well with dimensions
- Assume we can evaluate unnormalised $\tilde{p}(\mathbf{z})$
- At time τ maintain a record of the current state $\mathbf{z}^{(\tau)}$
- Make proposal conditioned on current state $\mathbf{z} = \mathbf{z}^{(\tau)}$, $q(\mathbf{z}' | \mathbf{z})$
- Symmetric proposal $q(\mathbf{z}' | \mathbf{z}) = q(\mathbf{z} | \mathbf{z}')$
- Accept or reject samples with probability:

$$r(\mathbf{z}', \mathbf{z}) = \min \left(1, \frac{\tilde{p}(\mathbf{z}')}{\tilde{p}(\mathbf{z})} \right)$$

- Sequence of samples $\mathbf{z}^{(1)}, \mathbf{z}^{(2)}, \dots$, form a *Markov chain*

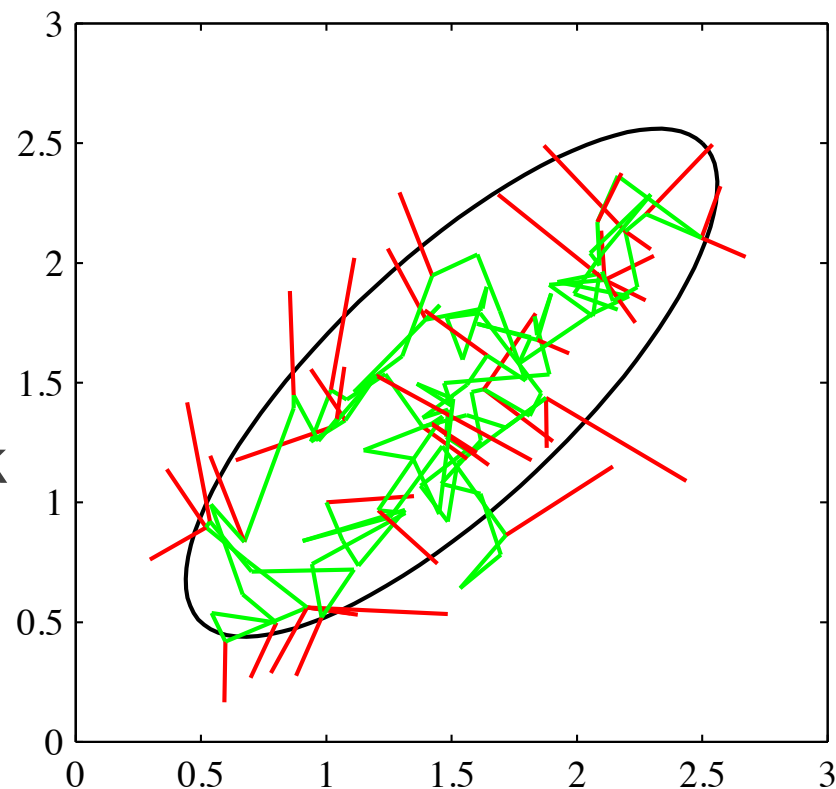
Basic Metropolis Algorithm (2)

1. Let $\mathbf{z} = \mathbf{z}^{(\tau)}$
2. Generate sample $\mathbf{z}' \sim q(\mathbf{z}'|\mathbf{z})$
3. Compute acceptance probability $r = \min \left(1, \frac{\tilde{p}(\mathbf{z}')}{\tilde{p}(\mathbf{z})} \right)$
4. Draw $u \sim \text{Uniform}(0, 1)$
5. If $u < r$
 $\mathbf{z}^{(\tau+1)} = \mathbf{z}'$
else
 $\mathbf{z}^{(\tau+1)} = \mathbf{z}^{(\tau)}$

- If step increases $p(\mathbf{z})$, candidate is kept
- If $q(\mathbf{z}' | \mathbf{z})$ is positive, distribution of $\mathbf{z}^{(\tau)}$ tends to $p(\mathbf{z})$ as $\tau \rightarrow \infty$
- Note samples not independent
 - » Retain every Mth sample and discard the rest

An Illustration of the Metropolis Algorithm

- $p(\mathbf{z})$: Correlated Gaussian
- $q(\mathbf{z})$: Isotropic Gaussian
- 1-std dev contour of $p(\mathbf{z})$ in black
- Accepted steps in green
- Rejected steps in red
- 43/150 samples rejected



Some Properties of Markov Chains

First-order Markov Chains (1)

$$p(\mathbf{z}_{1:M}) = \boxed{p(\mathbf{z}^{(1)})} \prod_{m=1}^{M-1} \boxed{p(\mathbf{z}^{(m+1)} | \mathbf{z}^{(m)})}$$

Conditional independence assumptions?



Fully specified by **Initial distribution**

Transition probabilities

$$T_m(\mathbf{z}^{(m)}, \mathbf{z}^{(m+1)})$$

- Homogeneous: All $T_m(\mathbf{z}^{(m)}, \mathbf{z}^{(m+1)})$ are the same $T(\mathbf{z}^{(m)}, \mathbf{z}^{(m+1)})$
- Marginal: $p(\mathbf{z}^{(m+1)}) = \sum_{\mathbf{z}^{(m)}} T(\mathbf{z}^{(m)}, \mathbf{z}^{(m+1)}) p(\mathbf{z}^{(m)})$
- **Invariant or stationary distribution** wrt Markov chain
 - Each step in the chain leaves the distribution invariant

$$p^*(\mathbf{z}) = \sum_{\mathbf{z}'} T(\mathbf{z}', \mathbf{z}) p^*(\mathbf{z}')$$

First-order Markov Chains (2)

- **Detailed balance:** $p^*(\mathbf{z})T(\mathbf{z}, \mathbf{z}') = p^*(\mathbf{z}')T(\mathbf{z}', \mathbf{z})$

- Sufficient (but not necessary) condition for $p^*(\mathbf{z})$ to be invariant

- Proof:

$$\sum_{\mathbf{z}'} p^*(\mathbf{z}')T(\mathbf{z}', \mathbf{z}) = \sum_{\mathbf{z}'} p^*(\mathbf{z})T(\mathbf{z}, \mathbf{z}') = p^*(\mathbf{z}') \sum_{\mathbf{z}'} p(\mathbf{z}'|\mathbf{z}) = p^*(\mathbf{z})$$

- Reversible Markov chain: Respects detailed balance

- **Goal:** Use Markov chains to sample from a distribution
- Set up Markov chain such as the desired distribution is invariant
- Also require **ergodicity**: as $m \rightarrow \infty$, $p(\mathbf{z}^{(m)})$ converges to $p^*(\mathbf{z})$
 - » Invariant distribution called equilibrium distribution
 - » an ergodic Markov chain can have only one equilibrium distribution

How to construct moves/transitions/proposals?

Metropolis-Hastings

The Metropolis-Hastings (MH) Algorithm

- Generalisation of Metropolis to non-symmetric proposals
- Accept or reject samples with probability:

$$r(\mathbf{z}', \mathbf{z}) = \min \left(1, \frac{\tilde{p}(\mathbf{z}')/q(\mathbf{z}'|\mathbf{z})}{\tilde{p}(\mathbf{z})/q(\mathbf{z}|\mathbf{z}')} \right) = \min \left(1, \frac{\tilde{p}(\mathbf{z}')q(\mathbf{z}|\mathbf{z}')}{\tilde{p}(\mathbf{z})q(\mathbf{z}'|\mathbf{z})} \right)$$

– Reduces to Metropolis for symmetric proposals

- *MH transitions satisfy detailed balance*
- Proposal distributions
 - Critical for the performance of the algorithm
 - Admissible: Gives non-zero probability of moving to states that have non-zero probability in the target
 - Gaussian centred at the current state (Random-walk MH)
 - Gaussian independent of current state (independence sampler)

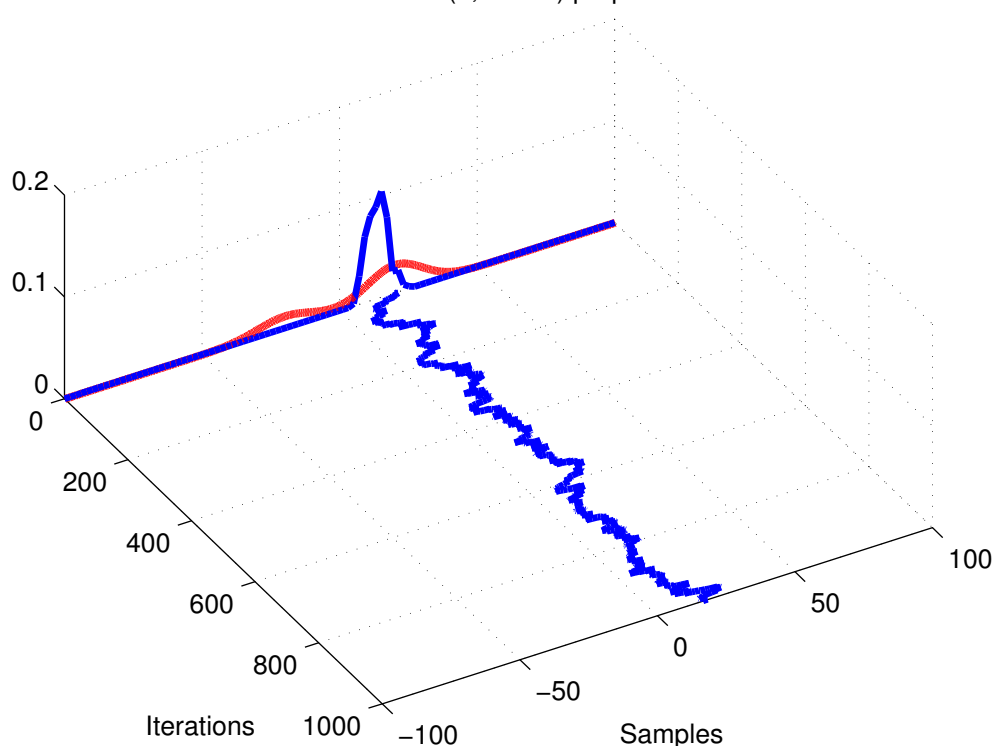
MH in Action (1)

- **Target distribution: Mixture of 1D-Gaussians**

- $\mu=(-20,20)$, $\pi=(0.3,0.7)$, $\sigma=(100,100)$

- Gaussian proposal $q(x'|x) = N(x' | x, \sigma^2)$

MH with $N(0,1.000^2)$ proposal



- Small variance $\sigma^2=1$
- Not much exploration
- Chain trapped near starting state
- Fails at sampling from one of the modes

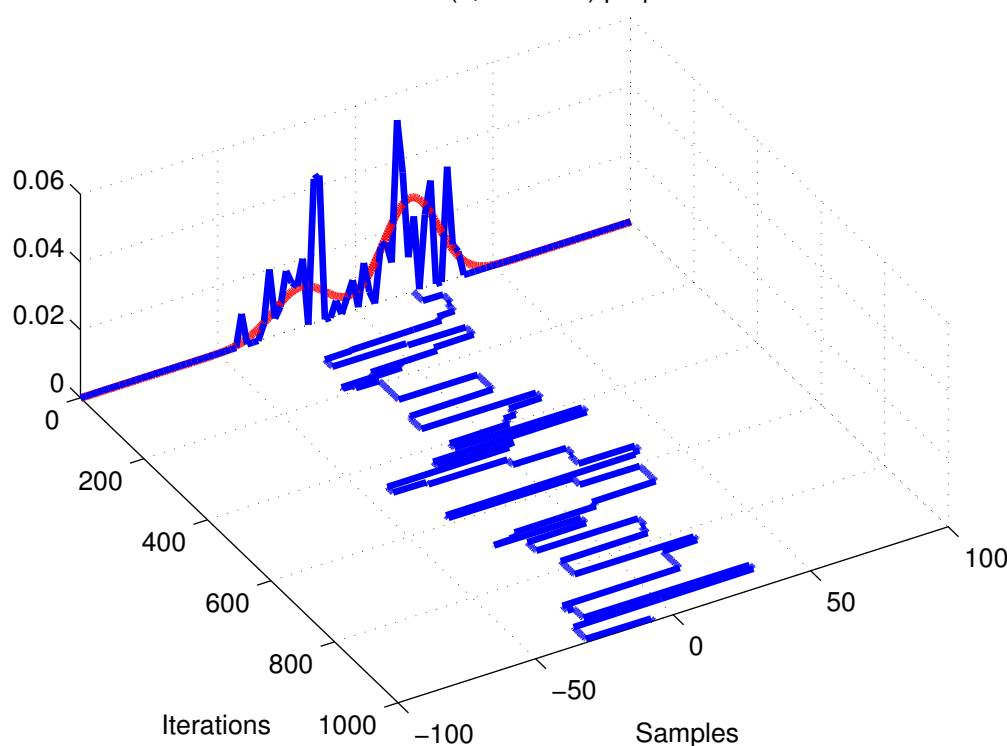
MH in Action (2)

- Target distribution: Mixture of 1D-Gaussians

- $\mu=(-20,20)$, $\pi=(0.3,0.7)$, $\sigma=(100,100)$

- Gaussian proposal $q(x'|x) = N(x' | x, \sigma^2)$

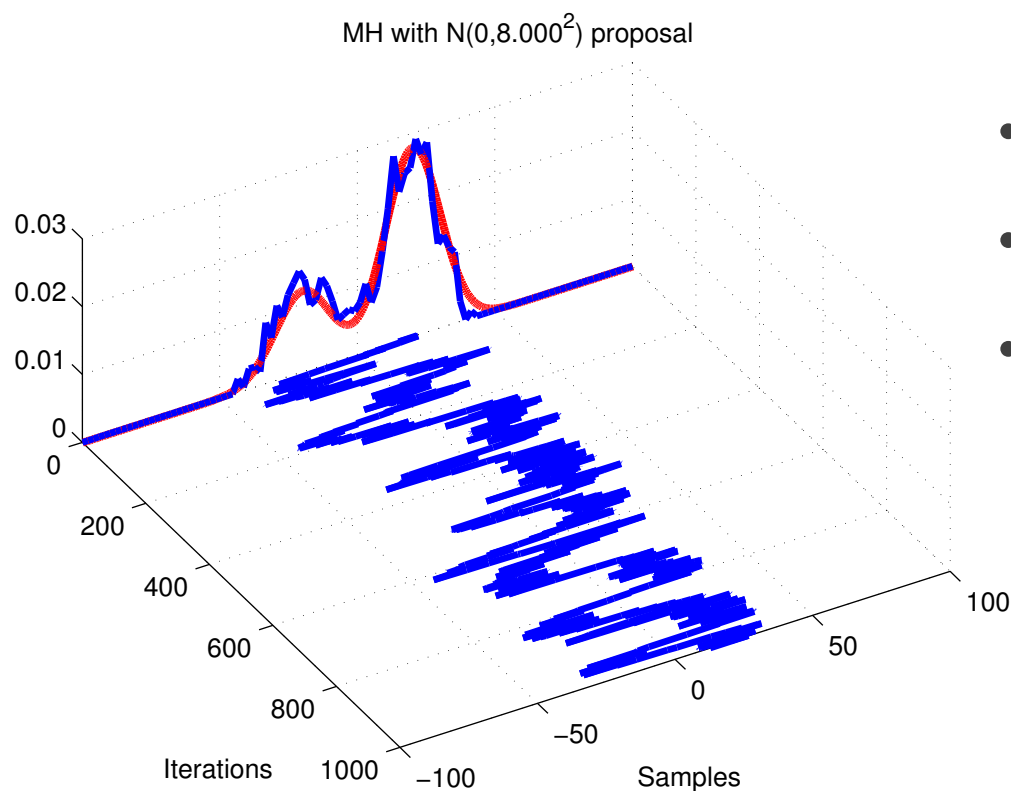
MH with $N(0,500.000^2)$ proposal



- Large variance $\sigma^2=500^2$
- High rejection rate
- **Sticky chain:** will stay in the same state for long
- Effective sample size is low (rough histogram in plot)

MH in Action (3)

- **Target distribution: Mixture of 1D-Gaussians**
 - $\mu=(-20,20)$, $\pi=(0.3,0.7)$, $\sigma=(100,100)$
- Gaussian proposal $q(x'|x) = N(x' | x, \sigma^2)$



- Right variance $\sigma^2=8^2$
- Explores both modes
- Good approximation of the true distribution

Gibbs Sampling

Gibbs Sampling

Goal: Sample from $p(\mathbf{z}) = p(z_1, z_2, \dots, z_M)$

- Main idea: Sample one variable at a time from its conditional distribution given all the other variables

1. Initialize $\{z_i : i = 1, \dots, M\}$

2. For $\tau = 1, \dots, T$

- Sample $z_1^{(\tau+1)} \sim p(z_1 | z_2^{(\tau)}, z_3^{(\tau)}, \dots, z_M^{(\tau)})$

- Sample $z_2^{(\tau+1)} \sim p(z_2 | z_1^{(\tau+1)}, z_3^{(\tau)}, \dots, z_M^{(\tau)})$

- Sample $z_j^{(\tau+1)} \sim p(z_j | z_1^{(\tau+1)}, \dots, z_{j-1}^{(\tau+1)}, z_{j+1}^{(\tau)}, \dots, z_M^{(\tau)})$

- Sample $z_M^{(\tau+1)} \sim p(z_M | z_1^{(\tau+1)}, z_2^{(\tau+1)}, \dots, z_{M-1}^{(\tau+1)})$

Why Gibbs Samples from Correct $p(\mathbf{z})$?

Joint distribution is invariant under each step

- Under update $z_i \sim p(z_i | \mathbf{z}_{\setminus i})$ marginal $p(\mathbf{z}_{\setminus i})$ is invariant as none of $\mathbf{z}_{\setminus i}$ gets updated
- Each step by definition samples from the correct conditional $p(z_i | \mathbf{z}_{\setminus i})$

Sufficient condition for ergodicity:

- None of the conditional distributions is anywhere zero
 - Any point can be reached from any other point in a finite number of steps
 - If this does not apply, ergodicity must be proven explicitly
- Note successive samples are highly correlated

Gibbs Sampling is a MH algorithm

- Use the proposal:

$$q(\mathbf{z}'|\mathbf{z}) = p(\mathbf{z}'_i|\mathbf{z}_{\setminus i})\mathbb{I}(\mathbf{z}'_{\setminus i} = \mathbf{z}_{\setminus i})$$

– z_i is sampled from full conditional and $\mathbf{z}_{\setminus i}$ remains unchanged

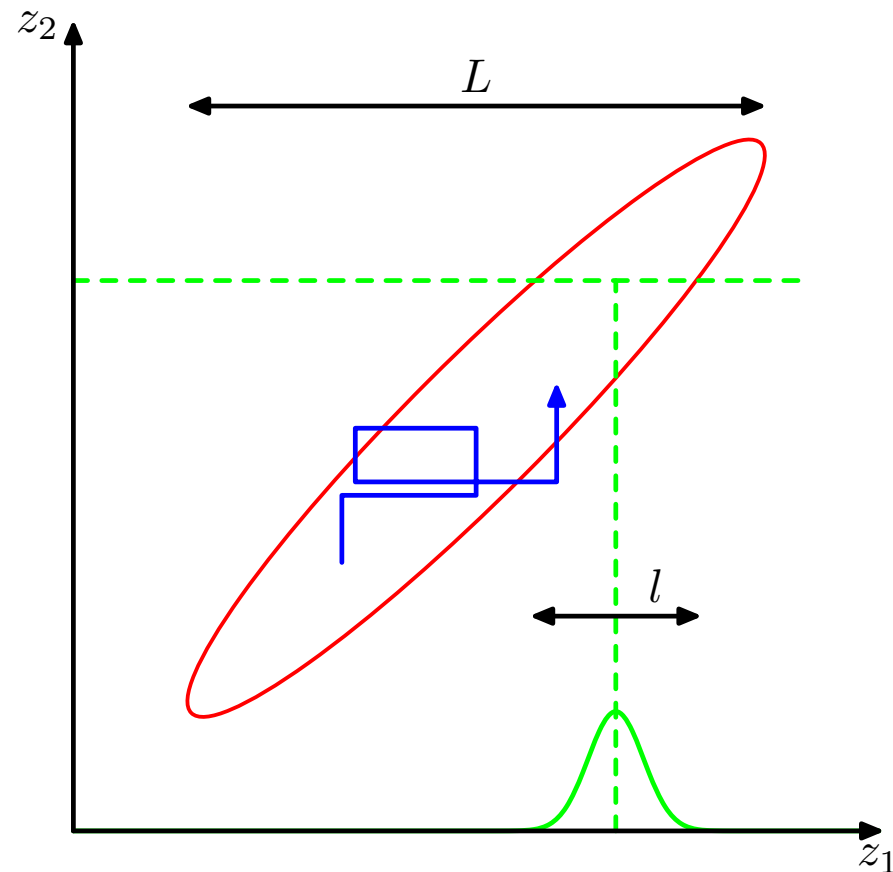
- Acceptance rate:

$$\begin{aligned}\frac{p(\mathbf{z}')q(\mathbf{z}|\mathbf{z}')}{p(\mathbf{z})q(\mathbf{z}'|\mathbf{z})} &= \frac{p(z'_i|\mathbf{z}'_{\setminus i})p(\mathbf{z}'_{\setminus i})p(z_i|\mathbf{z}'_{\setminus i})}{p(z_i|\mathbf{z}_{\setminus i})p(\mathbf{z}_{\setminus i})p(z'_i|\mathbf{z}_{\setminus i})} \quad \text{but } \mathbf{z}'_{\setminus i} = \mathbf{z}_{\setminus i} \\ &= \frac{p(z'_i|\mathbf{z}_{\setminus i})p(\mathbf{z}_{\setminus i})p(z_i|\mathbf{z}_{\setminus i})}{p(z_i|\mathbf{z}_{\setminus i})p(\mathbf{z}_{\setminus i})p(z'_i|\mathbf{z}_{\setminus i})} = 1\end{aligned}$$

100% acceptance rate does not necessarily mean rapid convergence (only one variable updated at a time)

Sampling from Gaussian using Gibbs

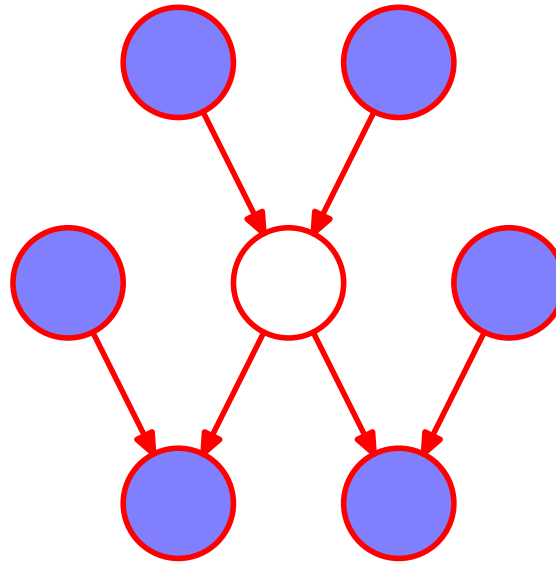
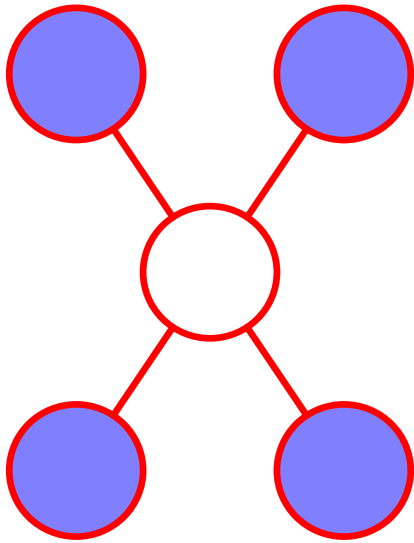
- 2D correlated Gaussian
- Gibbs updates only one variable at a time
- Marginal variance L
- Conditional variance l
- Step size governed by l
- # steps to produce independent sample $\mathcal{O}((L/l)^2)$



Gibbs can be very slow for correlated variables

Gibbs for Graphical Models

- Practical use depends on easiness of drawing from conditionals
 - Function of node states in Markov blanket



Can combine with adaptive rejection sampling techniques

- In general, strong dependencies between successive samples
 - Sample from groups of variables → Blocked Gibbs sampling

Practical Considerations of Sampling

- Burn-in phase

- MCMC starts from arbitrary state and only when the chain has “forgotten” where it started we will get samples from stationary distribution
 - » Ignore samples from initial period

- Mixing times

- Amount of time for a Markov chain to converge to stationary distribution
- Distributions with well-separated modes have high mixing times
- Standard MCMC methods do not work well in these cases

- Convergence diagnostics

- Heuristics for assessing non-convergence
- Trace plot: run multiple chains from very different overdispersed starting points, and plot samples of some variables of interest
- Compare variance of a quantity within chain vs variance across chains
 - » Estimated potential scale reduction

Conclusions

- Stochastic approaches to probabilistic inference
 - Sampling from posterior distribution
 - Expectations over functions
- Rejection Sampling and Importance Sampling
 - Useful in low-dimensional spaces
- Markov Chain Monte Carlo
 - Theory of Markov chains
 - Exact in the limit of infinite resources
 - Generic and easy to implement
 - Practical issues with convergence and diagnostics
- Reading
 - Bishop (PRML, 2006): Ch. 11 (except Sec. 11.4, 11.5, 11.6)
 - Murphy (MLaPP, 2012): Sec. 2.6, 2.7, Ch. 23 (except 23.5, 23.6), Ch.24 (except Sec. 24.5, 24.6, 24.7)